Chapter 18

Neural Control and Coordination

The functions of the organs and organ systems in our body must be coordinated to maintain homeostasis, coordination being the process through which two or more organs interact and complement the functions of one another.

What physical exercise shows

- when we do physical exercises the energy demand rises to keep up an increased muscular activity, and the supply of oxygen rises with it
- the increased supply of oxygen calls for an increase in the rate of respiration, in heart beat and in blood flow through the blood vessels
- when the exercise is stopped, the activities of nerves, lungs, heart and kidney gradually return to their normal conditions
- so the functions of muscles, lungs, heart, blood vessels, kidney and other organs are coordinated all through

The two systems that do it

- in our body the neural system and the endocrine system jointly coordinate and integrate all the activities of the organs, so that they function in a synchronised fashion
- the neural system provides an organised network of point-to-point connections for a quick coordination
- the endocrine system provides chemical integration through hormones

What this chapter covers

- the neural system of human
- mechanisms of neural coordination such as the transmission of a nerve impulse
- impulse conduction across a synapse

In this chapter

18.1 The neural system

18.2 The human neural system

18.3 The neuron

18.6 Conduction of the nerve impulse

18.7 Transmission of impulses

18.8 The central neural system

18.1 The neural system

Simple networks to developed systems

- the neural system of all animals is composed of highly specialised cells called neurons, which can detect, receive and transmit different kinds of stimuli
- the neural organisation is very simple in lower invertebrates, and in Hydra, for example, it is composed of a network of neurons
- it is better organised in insects, where a brain is present along with a number of ganglia and neural tissues
- the vertebrates have a more developed neural system

18.2 The human neural system

CNS and PNS

- the human neural system is divided into two parts, the central neural system (CNS) and the peripheral neural system (PNS)
- the CNS includes the brain and the spinal cord and is the site of information processing and control
- the PNS comprises all the nerves of the body associated with the CNS
- the nerve fibres of the PNS are of two types, afferent and efferent, the afferent fibres carrying impulses from tissues and organs to the CNS and the efferent fibres carrying regulatory impulses from the CNS to the peripheral tissues and organs concerned

Divisions of the PNS

- the PNS is divided into the somatic neural system and the autonomic neural system
- the somatic neural system relays impulses from the CNS to skeletal muscles, while the autonomic neural system carries impulses from the CNS to the involuntary organs and the smooth muscles of the body
- the autonomic neural system is further classified into the sympathetic neural system and the parasympathetic neural system
- the visceral nervous system is the part of the peripheral nervous system comprising the whole complex of nerves, fibres, ganglia and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system

18.3 The neuron

Cell body and dendrites

- a neuron is a microscopic structure composed of three major parts, the cell body, the dendrites and the axon
- the cell body contains cytoplasm with typical cell organelles and certain granular bodies called Nissl's granules
- short fibres which branch repeatedly and project out of the cell body also contain Nissl's granules and are called dendrites

Cell body, with its cytoplasm and Nissl's granules

Dendrites, the short branching fibres

Axon, the long fibre leaving the cell body

Myelin sheath from the Schwann cells, broken at the nodes of Ranvier

Synaptic knobs, the bulb-like endings of the axon branches

Figure 18.A Structure of a neuron

18.4 Kinds of neuron and axon

Which way an impulse travels

- dendrites transmit impulses towards the cell body
- the axon is a long fibre whose distal end is branched, each branch ending as a bulb-like structure called a synaptic knob, which holds synaptic vesicles containing chemicals called neurotransmitters
- axons transmit nerve impulses away from the cell body to a synapse or to a neuro-muscular junction

Three types of neuron

- neurons are divided by the number of axons and dendrites into three types
- multipolar, with one axon and two or more dendrites, found in the cerebral cortex
- bipolar, with one axon and one dendrite, found in the retina of the eye
- unipolar, a cell body with one axon only, found usually in the embryonic stage

Two types of axon

- the myelinated nerve fibres are enveloped with Schwann cells, which form a myelin sheath around the axon, and the gaps between two adjacent myelin sheaths are called nodes of Ranvier
- myelinated nerve fibres are found in spinal and cranial nerves
- an unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon, and is commonly found in the autonomous and the somatic neural systems

18.5 The resting state of the membrane

Why a neuron is polarised

- neurons are excitable cells because their membranes are in a polarised state
- different types of ion channels are present on the neural membrane, each selectively permeable to different ions
- when a neuron is resting, the axonal membrane is comparatively more permeable to potassium ions (K+) and nearly impermeable to sodium ions (Na+), and it is impermeable as well to the negatively charged proteins present in the axoplasm
- so the axoplasm holds a high concentration of K+ and of negatively charged proteins and a low concentration of Na+, while the fluid outside the axon holds a low concentration of K+ and a high concentration of Na+, which together form a concentration gradient

Point A, where the stimulus is applied and Na+ rushes in

Reversed polarity at A, negative outside and positive inside

Point B, still positive outside and negative inside

Current flowing inside from A to B and outside from B to A

Figure 18.B Diagrammatic representation of impulse conduction through an axon (at points A and B)

18.6 Generation and conduction of the nerve impulse

The resting potential

- these ionic gradients across the resting membrane are maintained by the active transport of ions by the sodium-potassium pump, which carries 3 Na+ outwards for 2 K+ into the cell
- as a result the outer surface of the axonal membrane carries a positive charge while its inner surface becomes negatively charged, and the membrane is therefore polarised
- the electrical potential difference across the resting plasma membrane is called the resting potential

Depolarisation at the point of stimulus

- when a stimulus is applied at a site on the polarised membrane, say point A, the membrane there becomes freely permeable to Na+
- this leads to a rapid influx of Na+ followed by a reversal of polarity at that site, so the outer surface becomes negatively charged and the inner side positively charged
- the polarity at site A is thus reversed and the membrane there is depolarised
- the electrical potential difference across the plasma membrane at site A is called the action potential, which is in fact the nerve impulse

How the impulse travels on

- at sites immediately ahead, say site B, the axon membrane still has a positive charge outside and a negative charge inside
- so a current flows on the inner surface from site A to site B, and on the outer surface from site B to site A, completing the circuit of current flow
- the polarity at that site is thereby reversed and an action potential is generated at site B, so the impulse generated at A arrives at B, and the sequence repeats along the length of the axon until the impulse is conducted
- the stimulus-induced rise in permeability to Na+ is extremely short-lived and is quickly followed by a rise in permeability to K+, and within a fraction of a second K+ diffuses outside the membrane, restores the resting potential at the site of excitation and leaves the fibre responsive once more to further stimulation

Axon terminal, holding the synaptic vesicles

Synaptic vesicles, filled with neurotransmitter

Synaptic cleft, the fluid-filled space between the two membranes

Post-synaptic membrane, carrying the receptors

Figure 18.C Diagram showing axon terminal and synapse

18.7 Transmission of impulses

What a synapse is

- a nerve impulse is transmitted from one neuron to another through junctions called synapses
- a synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a gap called the synaptic cleft
- there are two types, electrical synapses and chemical synapses

The electrical synapse

- at an electrical synapse the membranes of the pre- and post-synaptic neurons lie in very close proximity
- electrical current can flow directly from one neuron into the other across these synapses, and transmission here is very similar to impulse conduction along a single axon
- transmission across an electrical synapse is always faster than across a chemical synapse
- electrical synapses are rare in our system

The chemical synapse

- at a chemical synapse the membranes of the pre- and post-synaptic neurons are separated by a fluid-filled space, the synaptic cleft
- chemicals called neurotransmitters carry out the transmission of impulses at these synapses, and the axon terminals hold vesicles filled with them
- when an impulse arrives at the axon terminal it stimulates the movement of the synaptic vesicles towards the membrane, where they fuse with the plasma membrane

Forebrain, the cerebrum with the thalamus and hypothalamus

Midbrain, between the forebrain and the pons

Hindbrain, the pons, cerebellum and medulla

Cranial meninges, dura mater, arachnoid and pia mater, around the whole

Figure 18.D Diagram showing sagital section of the human brain

Release at the cleft

- the vesicles release their neurotransmitters into the synaptic cleft
- the released neurotransmitters bind to their specific receptors present on the post-synaptic membrane
- this binding opens ion channels, allowing the entry of ions that can generate a new potential in the post-synaptic neuron
- the new potential developed may be either excitatory or inhibitory

18.8 The central neural system

What the brain does

- the brain is the central information processing organ of our body and acts as the command and control system
- it controls voluntary movements, the balance of the body, the functioning of vital involuntary organs such as the lungs, heart and kidneys, thermoregulation, hunger and thirst, the circadian, that is 24-hour, rhythms of our body, the activities of several endocrine glands and human behaviour
- it is also the site for the processing of vision, hearing, speech, memory, intelligence, emotions and thoughts

How it is protected and divided

- the human brain is well protected by the skull
- inside the skull it is covered by the cranial meninges, an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer in contact with the brain tissue called pia mater
- the brain can be divided into three major parts, the forebrain, the midbrain and the hindbrain

18.9 The forebrain

The cerebrum

- the forebrain consists of the cerebrum, the thalamus and the hypothalamus, the cerebrum forming the major part of the human brain
- a deep cleft divides the cerebrum longitudinally into two halves, the left and right cerebral hemispheres, connected by a tract of nerve fibres called the corpus callosum
- the layer of cells covering the cerebral hemisphere is the cerebral cortex, thrown into prominent folds, and it is referred to as the grey matter because of its greyish appearance, which comes from the neuron cell bodies concentrated there
- fibres of the tracts are covered with the myelin sheath and constitute the inner part of the cerebral hemisphere, giving that layer an opaque white appearance, hence the white matter

Cortex, thalamus, hypothalamus

- the cerebral cortex contains motor areas, sensory areas and large regions that are neither clearly sensory nor motor, and these association areas are responsible for complex functions such as intersensory associations, memory and communication
- the cerebrum wraps around a structure called the thalamus, a major coordinating centre for sensory and motor signaling
- another very important part, the hypothalamus, lies at the base of the thalamus, and it contains a number of centres which control body temperature and the urge for eating and drinking
- it also contains several groups of neurosecretory cells, which secrete hormones called hypothalamic hormones

The limbic system

- the inner parts of the cerebral hemispheres, together with a group of associated deep structures such as the amygdala and the hippocampus, form a complex structure called the limbic lobe or limbic system
- along with the hypothalamus it is involved in the regulation of sexual behaviour, in the expression of emotional reactions such as excitement, pleasure, rage and fear, and in motivation

18.10 Midbrain and hindbrain

The midbrain

- the midbrain lies between the thalamus and hypothalamus of the forebrain and the pons of the hindbrain
- a canal called the cerebral aqueduct passess through it
- its dorsal portion consists mainly of four round swellings or lobes called corpora quadrigemina

The hindbrain and the brain stem

- the hindbrain comprises the pons, the cerebellum and the medulla, also called the medulla oblongata
- the pons consists of fibre tracts that interconnect different regions of the brain, and the cerebellum has a very convoluted surface in order to provide the additional space for many more neurons
- the medulla of the brain is connected to the spinal cord and contains centres which control respiration, cardiovascular reflexes and gastric secretions
- three major regions make up the brain stem, the mid brain, the pons and the medulla oblongata, and the brain stem forms the connections between the brain and the spinal cord

18.11 Summary

Neurons, impulses and synapses

- the neural system coordinates and integrates the functions as well as the metabolic and homeostatic activities of all the organs
- neurons, the functional units of the neural system, are excitable cells because of a differential concentration gradient of ions across the membrane, and the electrical potential difference across the resting neural membrane is the resting potential
- the nerve impulse is conducted along the axon membrane as a wave of depolarisation and repolarisation
- a synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a gap called the synaptic cleft, and the chemicals involved in transmission at chemical synapses are called neurotransmitters

The parts of the system

- the human neural system consists of the central neural system (CNS) and the peripheral neural system, the CNS being the brain and the spiral cord
- the brain divides into forebrain, midbrain and hindbrain, the forebrain consisting of cerebrum, thalamus and hypothalamus, with the cerebrum longitudinally divided into two halves connected by the corpus callosum
- the hypothalamus controls body temperature, eating and drinking, while the inner parts of the cerebral hemispheres with a group of associated deep structures form the limbic system, concerned with olfaction, autonomic responses, regulation of sexual behaviour, expression of emotional reactions and motivation
- the midbrain receives and integrates visual, tactile and auditory inputs, and the hindbrain comprises pons, cerebellum and medulla, the cerebellum integrating information received from the semicircular canals of the ear and the auditory system, the medulla holding centres that control respiration, cardiovascular reflexes and gastric secretions, and the pons consisting of fibre tracts that interconnect different regions of the brain

18.12 Exercises

From this chapter

1. Briefly describe the structure of the Brain
2. Compare the following: (a) Central neural system (CNS) and Peripheral neural system (PNS) (b) Resting potential and action potential
3. Explain the following processes: (a) Polarisation of the membrane of a nerve fibre (b) Depolarisation of the membrane of a nerve fibre (c) Transmission of a nerve impulse across a chemical synapse

18.13 Exercises, continued

From this chapter, continued

1. Draw labelled diagrams of the following: (a) Neuron (b) Brain
2. Write short notes on the following: (a) Neural coordination (b) Forebrain (c) Midbrain (d) Hindbrain (e) Synapse
3. Give a brief account of Mechanism of synaptic transmission.
4. Explain the role of Na+ in the generation of action potential.
5. Differentiate between: (a) Myelinated and non-myelinated axons (b) Dendrites and axons (c) Thalamus and Hypothalamus (d) Cerebrum and Cerebellum
6. Answer the following: (a) Which part of the human brain is the most developed? (b) Which part of our central neural system acts as a master clock?
7. Distinguish between: (a) afferent neurons and efferent neurons (b) impulse conduction in a myelinated nerve fibre and unmyelinated nerve fibre (f) cranial nerves and spinal nerves.